



2026

BIOLOGY

Grade 9 NCE | Paper 1

Time allowed: **45 minutes** Total marks: **50**

Candidate Name: _____

Index Number: _____

Class: _____

INSTRUCTIONS TO CANDIDATES

1. Write your name, index number and class in the spaces provided above.
2. Answer ALL questions in Sections A, B and C.
3. For Section A, place a tick (✓) in the box next to your chosen answer. Only one answer per question.
4. Write your answers in the spaces provided in Sections B and C.
5. Use black or dark blue ink for all written answers. Use pencil for diagrams only.
6. The number of marks available is shown in brackets [] at the end of each question or part question.
7. Do not use correction fluid. Cross out any work you do not want to be marked.

Topics covered: Cell Structure & Transport • Ecology & Nutrient Cycles

For Examiner's Use Only

Section A /15	Section B /20	Section C /15	Total /50
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SECTION A — Multiple Choice Questions (15 marks)

Place a tick (✓) in ONE box for each question. Each question carries 1 mark.

- 1 Which organelle is responsible for producing energy (ATP) through cellular respiration? [1]
- ☐ A Nucleus
- ☐ B Mitochondria
- ☐ C Ribosome
- ☐ D Chloroplast
- 2 Which of the following is found in plant cells but NOT in animal cells? [1]
- ☐ A Cell membrane
- ☐ B Mitochondria
- ☐ C Cell wall
- ☐ D Cytoplasm
- 3 Osmosis is best defined as: [1]
- ☐ A Movement of solute from high to low concentration
- ☐ B Movement of water molecules from a region of high water potential to low water potential through a semi-permeable membrane
- ☐ C Active movement of ions against a concentration gradient
- ☐ D Movement of gases from low to high pressure
- 4 Which process moves substances against a concentration gradient and requires energy? [1]
- ☐ A Diffusion
- ☐ B Osmosis
- ☐ C Active transport
- ☐ D Facilitated diffusion
- 5 Water is transported from roots to leaves through: [1]
- ☐ A Phloem vessels
- ☐ B Stomata
- ☐ C Lenticels
- ☐ D Xylem vessels
- 6 Sugars produced during photosynthesis are transported through the plant in the: [1]

- ☐ A Xylem
- ☐ B Phloem
- ☐ C Cortex
- ☐ D Epidermis

7 The loss of water vapour from leaves is called:

[1]

- ☐ A Transpiration
- ☐ B Respiration
- ☐ C Guttation
- ☐ D Evaporation

8 In a food chain: Grass → Grasshopper → Frog → Snake → Hawk. What is the grass?

[1]

- ☐ A Primary consumer
- ☐ B Secondary consumer
- ☐ C Producer
- ☐ D Decomposer

9 Which term describes all the living organisms in an area together with their non-living environment?

[1]

- ☐ A Community
- ☐ B Population
- ☐ C Ecosystem
- ☐ D Habitat

10 In the nitrogen cycle, which bacteria convert nitrates back to nitrogen gas?

[1]

- ☐ A Nitrifying bacteria
- ☐ B Nitrogen-fixing bacteria
- ☐ C Decomposers
- ☐ D Denitrifying bacteria

11 A cell placed in a hypertonic solution will:

[1]

- ☐ A Swell and burst
- ☐ B Remain unchanged
- ☐ C Shrink (crenate)
- ☐ D Divide by mitosis

12 Which of the following best describes a limiting factor in a population?

[1]

- ☐ A A factor that increases population growth

- ☐ B Any factor that restricts population size
- ☐ C The maximum size a population can reach
- ☐ D The rate at which organisms reproduce

13 Carbon is returned to the atmosphere mainly through:

[1]

- ☐ A Photosynthesis
- ☐ B Nitrogen fixation
- ☐ C Respiration and combustion
- ☐ D Transpiration

14 Root hair cells are adapted for absorption of water by:

[1]

- ☐ A Having a thick cell wall
- ☐ B Having a large surface area and thin cell wall
- ☐ C Containing many chloroplasts
- ☐ D Having no cell membrane

15 Which level of a food pyramid contains the MOST energy?

[1]

- ☐ A Top carnivores
- ☐ B Secondary consumers
- ☐ C Primary consumers
- ☐ D Producers

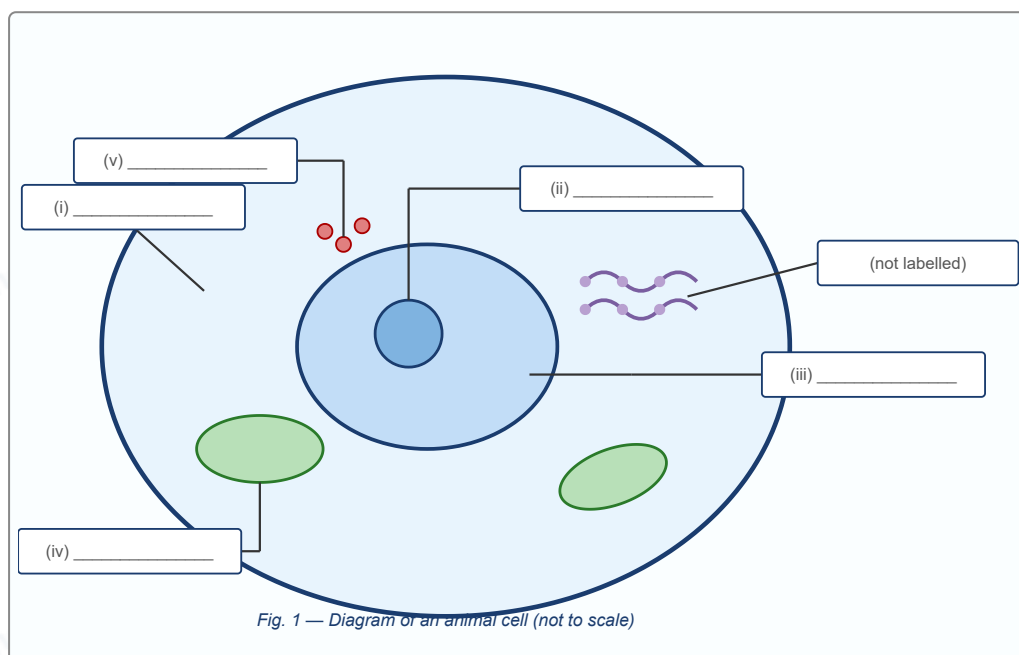
SECTION B — Structured Questions (20 marks)

Answer ALL questions. Write your answers in the spaces provided.

B1 Label the Animal Cell Diagram

[5]

The diagram below shows the structure of an animal cell. Write the correct label in each blank box provided. Choose from: *nucleus*, *mitochondria*, *cell membrane*, *ribosome*, *rough endoplasmic reticulum*, *cytoplasm*, *nucleolus*.



Label boxes (i) to (v) refer to the five arrowed structures. Write the correct biological name in each box.

B2 Osmosis Experiment

[5]

A student places potato cylinders of equal mass in three different solutions: distilled water, 0.5 mol/dm^3 sucrose solution and 1.0 mol/dm^3 sucrose solution. After 30 minutes, the mass of each cylinder is recorded and the percentage change in mass is calculated.

(a) State the **independent variable** in this experiment.

[1]

(b) Predict what will happen to the potato cylinder placed in **distilled water**. Explain your answer using the term *water potential*.

[2]

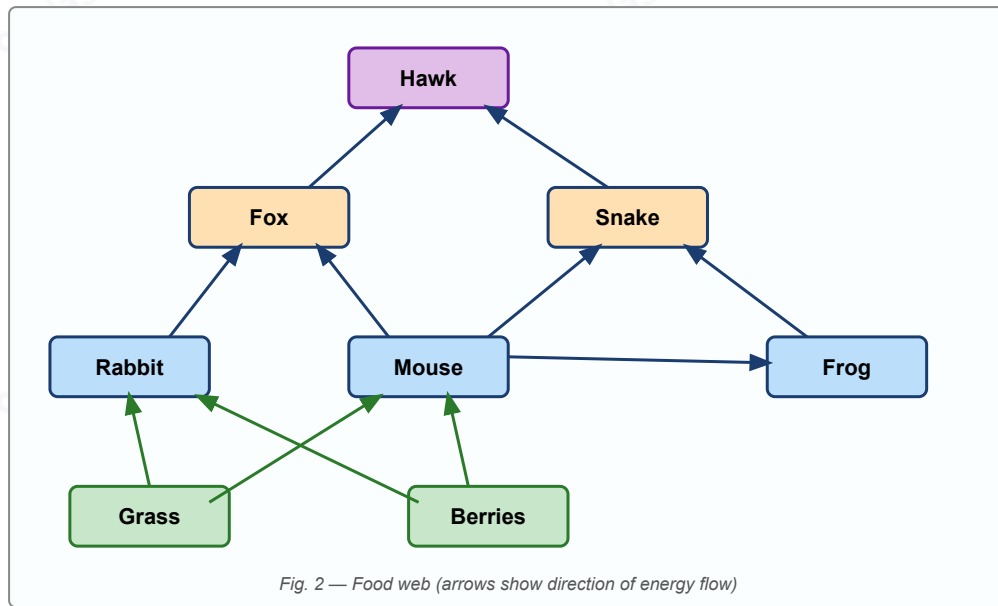
- (c) Explain why the potato cylinder placed in **1.0 mol/dm³ sucrose solution** loses mass.

[2]

B3 Food Web

[5]

Study the food web below and answer the questions that follow.



- (a) Name **one producer** and **one tertiary consumer** from the food web above. [2]

Producer: _____

Tertiary consumer: _____

- (b) What would happen to the **Hawk population** if all snakes were suddenly removed from this ecosystem? Explain your answer. [2]

- (c) State the **original source of energy** for this food web. [1]

B4 Transport in Plants

[5]

- (a) Name the **TWO types of vascular tissue** found in plants and state **one function** of each. [2]

Vascular Tissue	One Function

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- (b) Explain how water moves from the **soil into the root hair cell**. Name the process involved. [2]

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- (c) State **ONE factor** that would increase the rate of transpiration. [1]

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SECTION C — Extended Questions (15 marks)

Answer ALL questions. Write your answers in the spaces provided.

C1 Cell Structure and Function

[8]

- (a) Complete the table below to show whether each structure is present (✓) or absent (X) in each type of cell. [4]

Structure	Plant Cell	Animal Cell
Cell wall		
Chloroplast		
Large central vacuole		
Mitochondria		
Cell membrane		
Nucleus		

- (b) Explain how the structure of a **red blood cell** is adapted for its function of carrying oxygen. [2]

- (c) A student claims: *"All cells carry out the same functions."* Do you agree with this statement? Justify your answer with **TWO examples** of specialised cells and their specific functions. [2]

C2 Ecology — Energy and Nutrient Cycles

[7]

- (a) Explain what is meant by a *food web* and how it differs from a *food chain*.

[2]

- (b) Only about **10%** of the energy at one trophic level is transferred to the next. Explain where the other **90%** of energy goes.

[2]

- (c) Describe the role of **decomposers** in the carbon cycle.

[2]

- (d) State **ONE human activity** that disrupts the nitrogen cycle and explain its effect on the environment.

[1]

MARK SCHEME — Biology Paper 1 (2026 Predicted)

This mark scheme is intended for **teacher / tutor use only**. Accept any answer that conveys the correct biological meaning, unless a question specifies a precise term. Where alternative wording is acceptable, it is indicated with (*accept:...*).

SECTION A — Multiple Choice (1 mark each)

Q	Correct Answer	Marks
1	B — Mitochondria	1
2	C — Cell wall	1
3	B — Movement of water molecules from high water potential to low water potential through a semi-permeable membrane	1
4	C — Active transport	1
5	D — Xylem vessels	1
6	B — Phloem	1
7	A — Transpiration	1
8	C — Producer	1
9	C — Ecosystem	1
10	D — Denitrifying bacteria	1
11	C — Shrink (crenate)	1
12	B — Any factor that restricts population size	1
13	C — Respiration and combustion	1
14	B — Having a large surface area and thin cell wall	1
15	D — Producers	1
Section A Total		15

SECTION B — Structured Questions

Q	Answer / Marking Points	Marks
B1 (i)–(v)	(i) Cell membrane (ii) Nucleolus (iii) Nucleus (iv) Mitochondria (<i>accept: mitochondrion</i>) (v) Ribosome(s) <i>Award 1 mark for each correct label.</i>	5
B2 (a)	The concentration of sucrose solution (<i>accept: type/concentration of solution; solute concentration</i>)	1

B2 (b)	The potato cylinder will gain mass / increase in length (1). The water potential of the distilled water is higher than that of the potato cell sap, so water enters the potato cells by osmosis (1).	2
B2 (c)	The 1.0 mol/dm ³ sucrose solution has a lower water potential than the potato cells (1). Water moves out of the potato cells by osmosis into the sucrose solution, reducing the mass of the cylinder (1).	2
B3 (a)	Producer: Grass or Berries (1). Tertiary consumer: Hawk (1).	2
B3 (b)	The Hawk population would initially decrease (1) because one of its food sources (snake) has been removed, reducing the available prey (1). <i>Accept: the hawk would need to rely more on the fox, but if snake population goes to zero, energy supply for hawk falls.</i>	2
B3 (c)	The Sun (accept: sunlight / solar energy)	1
B4 (a)	Xylem — transports water (and mineral salts) from roots to leaves (1). Phloem — transports dissolved sugars (sucrose) / products of photosynthesis around the plant (1).	2
B4 (b)	Process: Osmosis (1). The soil water has a higher water potential than the cell sap in the root hair cell. Water moves across the partially permeable cell membrane from soil water into the root hair cell down the water potential gradient (1).	2
B4 (c)	Any ONE of: increased temperature / increased light intensity / increased wind speed / lower humidity / increased surface area of leaves. (1 mark)	1
Section B Total		20

SECTION C — Extended Questions

Q	Answer / Marking Points	Marks
C1 (a)	Award 1 mark for each correct row (any 4 correct rows = 4 marks; deduct where answer is clearly wrong): Cell wall: Plant ✓ / Animal X Chloroplast: Plant ✓ / Animal X Large central vacuole: Plant ✓ / Animal X Mitochondria: Plant ✓ / Animal ✓ Cell membrane: Plant ✓ / Animal ✓ Nucleus: Plant ✓ / Animal ✓ <i>Award 1 mark per correct pair (both columns must be correct for that row).</i>	4
C1 (b)	Red blood cells have a biconcave disc shape (1), which gives a large surface area to volume ratio for efficient diffusion of oxygen (1). They also lack a nucleus, leaving more space for	2

	haemoglobin. (<i>Accept any two structural adaptations with linked function; max 2 marks.</i>)	
C1 (c)	Disagree (no mark for agree/disagree alone). Cells are specialised for different functions (1). Example 1: Root hair cells — long extensions / large surface area for absorption of water and mineral salts (1). Example 2: Nerve cells (neurones) — long axons for rapid transmission of electrical impulses (1). <i>Accept any two valid specialised cells with correct function. Award 1 mark per correct example (name + function). Max 2 marks.</i>	2
C2 (a)	A food chain shows a single linear sequence of feeding relationships (1). A food web shows multiple interconnecting food chains, with organisms having more than one source of food / prey (1).	2
C2 (b)	The other 90% is lost as heat energy through cellular respiration (1). Energy is also used for movement, growth, and reproduction / some is lost in excretion and undigested material (faeces) (1).	2
C2 (c)	Decomposers (bacteria and fungi) break down dead organic matter / dead plants and animals (1). This releases carbon dioxide back into the atmosphere through respiration, completing the carbon cycle (1).	2
C2 (d)	Any ONE of: • Use of artificial fertilisers — excess nitrates leach into waterways causing eutrophication / algal blooms (1). • Burning fossil fuels — releases nitrogen oxides causing acid rain (1). • Deforestation — reduces nitrogen fixation, leads to nitrate loss from soil (1).	1
Section C Total		15
GRAND TOTAL		50